

Asignatura	Datos del alumno	Fecha
Programa Profesional en Inteligencia Artificial y Data Science	Apellidos: ORTIZ GARCIA	05/05/2025
	Nombre: ASIER	

Actividad 7 - HDFS

1. Introducción y entorno de trabajo

Aunque el ejercicio propone trabajar sobre un repositorio oficial preconfigurado he optado por no usar directamente ese entorno y construir una solución personalizada desde cero.

En lugar de utilizar las imágenes preconstruidas del repositorio original, he desarrollado un clúster propio en Docker a partir de una imagen base de Ubuntu 20.04, sobre la que he instalado manualmente Hadoop 2.7.4 y configurado todos los servicios requeridos (aunque no se usen todos): NameNode, DataNode, ResourceManager, NodeManager y HistoryServer.

Esto me ha permitido:

- Comprender mejor cómo se instala y configura Hadoop desde cero.
- Controlar de forma más precisa los scripts de arranque, el orden de los servicios y su interacción.
- Repasar conceptos de Docker, Makefile, shell scripting y configuración de entornos distribuidos que estudié anteriormente.

2. Pasos de la actividad

Paso 1

Para levantar todos los servicios del clúster, se ha utilizado el comando ``make start``, que construye las imágenes Docker si no existen y lanza los contenedores definidos en ``docker-compose.yml``.

Asignatura	Datos del alumno	Fecha
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```

asierortizgarcia@MacStudodeAsier:~/DocumentosMac/Git/ai-course-notebooks/actividad7
~/DocumentosMac/Git/ai-course-notebooks/actividad7 on main !1 make start base at 01:24:28
docker compose build
[+] Building 0.0s (0/1)                                docker:desktop-linux
[+] Building 0/1
[+] Building 3.8s (20/21)                                docker:desktop-linux
=> [namenode internal] load metadata for docker.io/library/ubuntu:20.04 3.3s
[+] Building 4.2s (38/47)                                docker:desktop-linux
=> [resource-manager internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
[+] Building 4.5s (61/85)                                docker:desktop-linux
=> [namenode internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 2.54kB 0.0s
=> [history-server internal] load metadata for docker.io/library/ubuntu:20.04 3.7s
=> [namenode auth] library/ubuntu:pull token for registry-1.docker.io 0.0s
=> [namenode internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [history-server 1/13] FROM docker.io/library/ubuntu:20.04@sha256:8feb4d8ca5354def3d8fce243717141ce31e2c4 0.0s
=> [namenode internal] load build context 0.0s
=> => transferring context: 7.06kB 0.0s
=> CACHED [actividad7 2/13] RUN apt-get update && apt-get install -y openjdk-8-jdk openssh-server 0.0s
=> CACHED [actividad7 3/13] RUN ln -s /usr/lib/jvm/java-8-openjdk-amd64 /usr/lib/jvm/default-java || true 0.0s
=> CACHED [actividad7 4/13] RUN wget https://archive.apache.org/dist/hadoop/core/hadoop-2.7.4/hadoop-2.7.4 0.0s
=> CACHED [actividad7 5/13] RUN mkdir -p /opt/hadoop-2.7.4/data/namenode /opt/hadoop-2.7.4/data/da 0.0s
=> [namenode 6/13] COPY hadoop-config/core-site.xml /opt/hadoop-2.7.4/etc/hadoop/core-site.xml 0.0s
=> [namenode 7/13] COPY hadoop-config/hdfs-site.xml /opt/hadoop-2.7.4/etc/hadoop/hdfs-site.xml 0.0s
=> [namenode 8/13] COPY hadoop-config/mapred-site.xml /opt/hadoop-2.7.4/etc/hadoop/mapred-site.xml 0.0s
=> [namenode 9/13] COPY hadoop-config/yarn-site.xml /opt/hadoop-2.7.4/etc/hadoop/yarn-site.xml 0.0s
=> [namenode 10/13] COPY entrypoint.sh /entrypoint.sh 0.0s
=> [namenode 11/13] COPY setup/actividad7.sh /setup/actividad7.sh 0.0s
=> [namenode 12/13] RUN chmod +x /entrypoint.sh /setup/actividad7.sh 0.1s
=> [namenode 13/13] RUN echo 'export HADOOP_HOME=/opt/hadoop-2.7.4' >> /root/.bashrc && echo 'export HA 0.1s
=> [namenode] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:083318cc5454d458fe48feae727c177458bcaae35b6703443946253d2ccee5f6 0.0s
=> => naming to docker.io/library/actividad7-namenode 0.0s
=> [namenode] resolving provenance for metadata file 0.0s
=> [resource-manager internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 2.54kB 0.0s
=> [datanode internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 2.54kB 0.0s
=> [datanode internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [resource-manager internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [resource-manager internal] load build context 0.0s
=> => transferring context: 618B 0.0s
=> [datanode internal] load build context 0.0s
=> => transferring context: 7.06kB 0.0s
=> CACHED [resource-manager 6/13] COPY hadoop-config/core-site.xml /opt/hadoop-2.7.4/etc/hadoop/core-site.x 0.0s
=> CACHED [resource-manager 7/13] COPY hadoop-config/hdfs-site.xml /opt/hadoop-2.7.4/etc/hadoop/hdfs-site.x 0.0s
=> CACHED [resource-manager 8/13] COPY hadoop-config/mapred-site.xml /opt/hadoop-2.7.4/etc/hadoop/mapred-si 0.0s
=> CACHED [resource-manager 9/13] COPY hadoop-config/yarn-site.xml /opt/hadoop-2.7.4/etc/hadoop/yarn-site.x 0.0s
=> CACHED [resource-manager 10/13] COPY entrypoint.sh /entrypoint.sh 0.0s
=> CACHED [resource-manager 11/13] COPY setup/actividad7.sh /setup/actividad7.sh 0.0s
=> CACHED [resource-manager 12/13] RUN chmod +x /entrypoint.sh /setup/actividad7.sh 0.0s
=> CACHED [datanode 13/13] RUN echo 'export HADOOP_HOME=/opt/hadoop-2.7.4' >> /root/.bashrc && echo 'ex 0.0s
=> [resource-manager] exporting to image 0.0s

```

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```

asierortizgarcia@MacStudodeAsier:~/DocumentosMac/Git/ai-course-notebooks/actividad7
=> [datanode] resolving provenance for metadata file 0.0s
=> [nodemanager internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 2.54kB 0.0s
=> [historyserver internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 2.54kB 0.0s
=> [actividad7 internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 2.54kB 0.0s
=> [historyserver internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [actividad7 internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [nodemanager internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [actividad7 internal] load build context 0.0s
=> => transferring context: 618B 0.0s
=> [historyserver internal] load build context 0.0s
=> => transferring context: 618B 0.0s
=> [nodemanager internal] load build context 0.0s
=> => transferring context: 7.06kB 0.0s
=> CACHED [nodemanager 13/13] RUN echo 'export HADOOP_HOME=/opt/hadoop-2.7.4' >> /root/.bashrc && echo 0.0s
=> [historyserver] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:dab4409378b35b21803e2bfdd4862b85e56572b13262d8c33f827a15416b96c6 0.0s
=> => naming to docker.io/library/actividad7-historyserver 0.0s
=> CACHED [actividad7 6/13] COPY hadoop-config/core-site.xml /opt/hadoop-2.7.4/etc/hadoop/core-site.xml 0.0s
=> CACHED [actividad7 7/13] COPY hadoop-config/hdfs-site.xml /opt/hadoop-2.7.4/etc/hadoop/hdfs-site.xml 0.0s
=> CACHED [actividad7 8/13] COPY hadoop-config/mapred-site.xml /opt/hadoop-2.7.4/etc/hadoop/mapred-site.xml 0.0s
=> CACHED [actividad7 9/13] COPY hadoop-config/yarn-site.xml /opt/hadoop-2.7.4/etc/hadoop/yarn-site.xml 0.0s
=> CACHED [actividad7 10/13] COPY entrypoint.sh /entrypoint.sh 0.0s
=> CACHED [actividad7 11/13] COPY setup/actividad7.sh /setup/actividad7.sh 0.0s
=> CACHED [actividad7 12/13] RUN chmod +x /entrypoint.sh /setup/actividad7.sh 0.0s
=> [nodemanager] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:35165787fc27e7d5bd1e9d1f1635e31a80308699063b13ef0bec19f3139b3425 0.0s
=> => naming to docker.io/library/actividad7-nodemanager 0.0s
[+] Building 6/6 exporting to image 0.0s
✔ Service namenode Built 3.8s
✔ Service resourcemanager Built 0.4s
✔ Service datanode Built 0.4s
✔ Service historyserver Built 0.4s
✔ Service nodemanager Built 0.4s
✔ Service actividad7 Built 0.4s
docker compose up -d
[+] Running 8/8
✔ Network actividad7_default Created 0.0s
✔ Volume "actividad7_hadoop_historyserver" Created 0.0s
✔ Container namenode Started 0.3s
✔ Container resourcemanager Started 0.4s
✔ Container datanode Started 0.4s
✔ Container historyserver Started 0.5s
✔ Container nodemanager Started 0.5s
✔ Container actividad7 Started 0.5s
Clúster levantado correctamente.
- NameNode UI: http://localhost:9870
- YARN UI: http://localhost:8088
- HistoryServer: http://localhost:19888

```

Durante el arranque, cada servicio ejecuta su propio comando a través del script `entrypoint.sh`, asegurando que el orden y la disponibilidad de HDFS se respeten antes de iniciar componentes como ResourceManager, NodeManager o HistoryServer.

Para comprobar que todos los servicios están correctamente levantados, se pueden usar los comandos `make logs`, `make stats` o acceder a sus interfaces web desde:

- NameNode UI: <http://localhost:9870>
- YARN ResourceManager: <http://localhost:8088>
- Job HistoryServer: <http://localhost:19888>

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	Nombre: ASIER	

make stats							
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
5e74c83b106b	namenode	0.49%	320.2MiB / 7.654GiB	4.09%	184kB / 45.1kB	24.6kB / 1.79MB	51
ee8a6bfdf7de	datanode	0.56%	273.6MiB / 7.654GiB	3.49%	35.5kB / 161kB	4.1kB / 803kB	52
9d692cd6ead1	resourcemanager	1.89%	422.6MiB / 7.654GiB	5.39%	285kB / 95.4kB	0B / 750kB	208
7adba3f4975a	nodemanager	1.66%	385.6MiB / 7.654GiB	4.92%	97.1kB / 283kB	0B / 696kB	91
4ee8251a1acc	historyserver	0.42%	376.9MiB / 7.654GiB	4.81%	6.42kB / 8.22kB	0B / 614kB	52

Namenode information

All Applications

JobHistory

localhost:50070/dfshealth.html#tab-overview

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Overview 'namenode:9000' (active)

Started:	Sun May 04 23:24:43 UTC 2025
Version:	2.7.4, rcd915e1e8d9d0131462a0b7301586c175728a282
Compiled:	2017-08-01T00:29Z by kshvachk from branch-2.7.4
Cluster ID:	CID-bdf6fe95-cc6c-4f30-a234-484c9d4dbda7
Block Pool ID:	BP-813063963-172.18.0.2-1746401080139

Summary

Security is off.

Safemode is off.

11 files and directories, 1 blocks = 12 total filesystem object(s).

Heap Memory used 143.53 MB of 203.5 MB Heap Memory. Max Heap Memory is 889 MB.

Non Heap Memory used 41.38 MB of 42.19 MB Committed Non Heap Memory. Max Non Heap Memory is -1 B.

Configured Capacity:	58.37 GB
DFS Used:	44 KB (0%)
Non DFS Used:	25.17 GB

Asignatura	Datos del alumno	Fecha
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	Nombre: ASIER	

Cluster

About NodesNode LabelsApplicationsNEWNEW_SAVINGSUBMITTEDACCEPTEDRUNNINGFINISHEDFAILEDKILLED

Scheduler

Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	VCores Used	VCores Total	VCores Reserved	Active Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes
0	0	0	0	0	0 B	8 GB	0 B	0	8	0	1	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation
Capacity Scheduler	[MEMORY]	<memory:1024, vCores:1>	<memory:8192, vCores:32>

Show 20 entries

ID	User	Name	Application Type	Queue	StartTime	FinishTime	State	FinalStatus	Progress	Tracking UI	Blacklisted Nodes
No data available in table											

Showing 0 to 0 of 0 entries

Application

About Jobs

Tools

Retired Jobs

Show 20 entries

Search:

Submit Time	Start Time	Finish Time	Job ID	Name	User	Queue	State	Maps Total	Maps Completed	Reduces Total	Reduces Completed
No data available in table											

Showing 0 to 0 of 0 entries

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Paso 2

Se crea el directorio `/books` en la raíz del sistema HDFS mediante `hdfs dfs -mkdir /books`.

```

root@dc50f5cbdc19: /
root@dc50f5cbdc19: /# hdfs dfs -mkdir /books
root@dc50f5cbdc19: /# hdfs dfs -ls /
Found 1 items
drwxr-xr-x  - root supergroup          0 2025-05-04 14:51 /books
root@dc50f5cbdc19: /#

```

Paso 3

Se crea un directorio personal bajo `/user/ubuntu/`: `/user/ubuntu/asier\_ortiz`.

```

root@dc50f5cbdc19: /
root@dc50f5cbdc19: /# hdfs dfs -mkdir -p /user/ubuntu/asier_ortiz
root@dc50f5cbdc19: /# hdfs dfs -ls /user/ubuntu
Found 1 items
drwxr-xr-x  - root supergroup          0 2025-05-04 14:56 /user/ubuntu/asier_ortiz
root@dc50f5cbdc19: /#

```

Paso 4

Se copian los archivos `frankenstein.txt` y `Quijote.txt` desde el sistema de archivos local (`/home/ubuntu/bigdata/examples/books`) al directorio `/books` en HDFS.

```

root@dc50f5cbdc19: /
root@dc50f5cbdc19: /# hdfs dfs -put /home/ubuntu/bigdata/examples/books/frankenstein.txt /books/
root@dc50f5cbdc19: /# hdfs dfs -put /home/ubuntu/bigdata/examples/books/Quijote.txt /books/
root@dc50f5cbdc19: /# hdfs dfs -ls /books
Found 2 items
-rw-r--r--  1 root supergroup    2141519 2025-05-04 14:59 /books/Quijote.txt
-rw-r--r--  1 root supergroup    441034 2025-05-04 14:58 /books/frankenstein.txt
root@dc50f5cbdc19: /#

```

Asignatura	Datos del alumno	Fecha
Programa Profesional en Inteligencia Artificial y Data Science	Apellidos: ORTIZ GARCIA	05/05/2025
	Nombre: ASIER	

Paso 5

Se descomprime el archivo `fichero\_result.zip` desde el directorio local `/home/ubuntu/bigdata/examples/hdfs/`, y se sube el archivo `fichero\_result.txt` extraído al directorio personal en HDFS.

```
root@dc50f5cbdc19: /
root@dc50f5cbdc19:/# unzip /home/ubuntu/bigdata/examples/hdfs/fichero_result.zip -d /tmp
Archive:  /home/ubuntu/bigdata/examples/hdfs/fichero_result.zip
  creating: /tmp/fichero_result/
  inflating: /tmp/_MACOSX/_fichero_result
  inflating: /tmp/fichero_result/fichero_result.txt
  inflating: /tmp/_MACOSX/fichero_result/_fichero_result.txt
root@dc50f5cbdc19:/# hdfs dfs -put /tmp/fichero_result/fichero_result.txt /user/ubuntu/asier_ortiz/
root@dc50f5cbdc19:/# hdfs dfs -ls /user/ubuntu/asier_ortiz/
Found 1 items
-rw-r--r--  1 root supergroup          4 2025-05-04 15:00 /user/ubuntu/asier_ortiz/fichero_result.txt
root@dc50f5cbdc19:/#
```

Paso 6

Se mueve el archivo `frankenstein.txt` del directorio `/books` al directorio del usuario `/user/ubuntu/asier\_ortiz`.

```
root@dc50f5cbdc19: /
root@dc50f5cbdc19:/# hdfs dfs -mv /books/frankenstein.txt /user/ubuntu/asier_ortiz/
root@dc50f5cbdc19:/# hdfs dfs -ls /user/ubuntu/asier_ortiz/
Found 2 items
-rw-r--r--  1 root supergroup          4 2025-05-04 15:00 /user/ubuntu/asier_ortiz/fichero_result.txt
-rw-r--r--  1 root supergroup    441034 2025-05-04 14:58 /user/ubuntu/asier_ortiz/frankenstein.txt
root@dc50f5cbdc19:/# hdfs dfs -ls /books
Found 1 items
-rw-r--r--  1 root supergroup    2141519 2025-05-04 14:59 /books/Quijote.txt
root@dc50f5cbdc19:/#
```

Paso 7

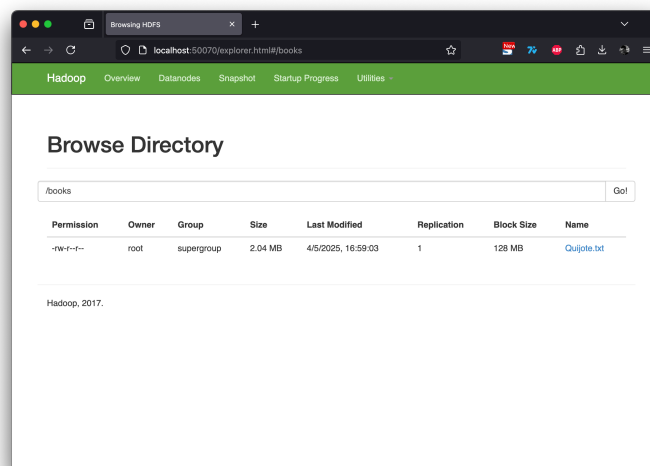
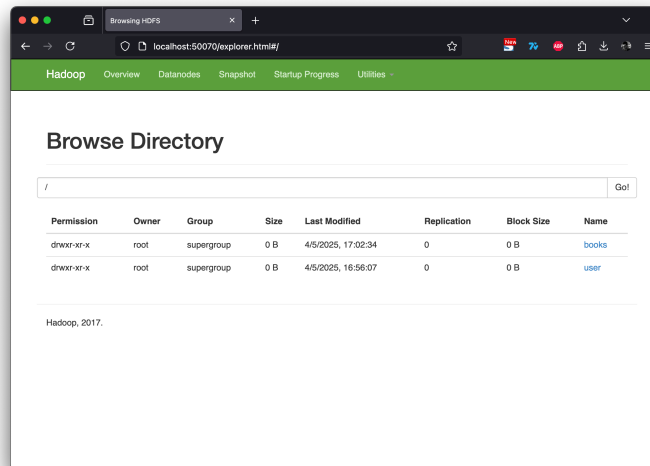
Se descarga el archivo `frankenstein.txt` desde HDFS a la ruta local `/home/ubuntu/bigdata/examples/hdfs/` para verificar que la operación fue exitosa.

```
root@dc50f5cbdc19: /
root@dc50f5cbdc19:/# hdfs dfs -get /user/ubuntu/asier_ortiz/frankenstein.txt /home/ubuntu/bigdata/examples/hdfs/
root@dc50f5cbdc19:/# ls -l /home/ubuntu/bigdata/examples/hdfs/
total 436
-rw-r--r--  1 root root   1151 May  4 13:17 fichero_result.zip
-rw-r--r--  1 root root  441034 May  4 15:05 frankenstein.txt
root@dc50f5cbdc19:/#
```

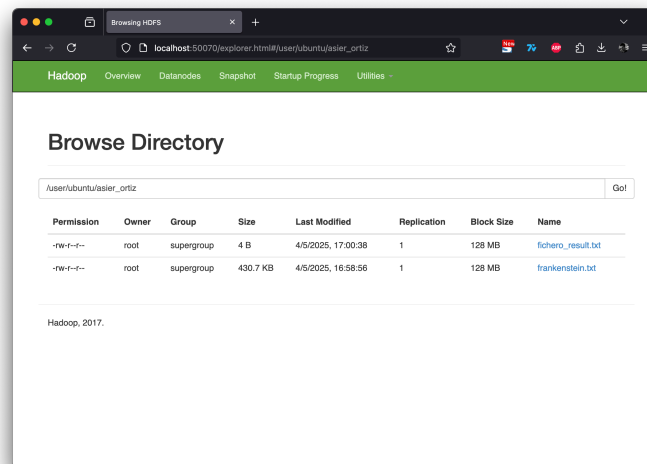
Asignatura	Datos del alumno	Fecha
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Paso 8

Se accede a la interfaz web del servicio NameNode para comprobar visualmente la estructura de HDFS.



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3. Automatización

Una vez comprobado que todos los pasos pueden realizarse correctamente de forma manual, se ha añadido un servicio extra en `docker-compose.yml` que ejecuta automáticamente el script `actividad7.sh`. Este script automatiza todas las operaciones de gestión de archivos que pide el ejercicio, incluyendo la creación de carpetas, carga de ficheros y movimientos dentro de HDFS.

```

actividad7:
  platform: ${PLATFORM}
  build:
    context: .
    dockerfile: Dockerfile
  container_name: actividad7
  command: /setup/actividad7.sh
  depends_on:
    - namenode
    - datanode
  volumes:
    - ./home/ubuntu/bigdata/examples:/home/ubuntu/bigdata/examples
  env_file:
    - .env

```

Todos los detalles adicionales sobre la estructura, configuración y uso de los comandos make definidos en el fichero Makefile están explicados en el README del repositorio:

<https://github.com/asier-ortiz/ai-course-notebooks/tree/main/actividad7>

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